# Practical: 15

Practical 15: Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

Problem Definition: Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.

## A. Theory

**1. Are all pages properly linked and navigable?**

Yes. All pages across static components (Home, Profile, Course, Attendance, Assignment, Results, Contact) and dynamic PHP modules (Practical 7 File Storage, Practical 8 DB, Practical 9 Registration, Practical 10 Auth & Sessions, Practical 11 Students, Practical 12 Events & Uploads, Practical 13 AJAX CRUD, Practical 14 Admin Dashboard, and Practical 15 Master Integration) are unified under a central, responsive navigation bar and sidebar with consistent relative routing.

**2. Are sessions, roles, validations, and database operations working end-to-end?**

Yes. User sessions utilize secure cookie parameters and automatic 30-minute inactivity expiration. Role-Based Access Control (RBAC) securely gates administrative controls (Audit logs, Student CRUD, Event management, System diagnostics) while permitting students to access event registrations and profiles. Dual-layer validation (HTML5/JS on frontend, regex/filter_var on backend) protects against invalid inputs, and all MySQLi queries execute through parameterized prepared statements.

**3. Are security and accessibility checklists completed?**

Yes. Comprehensive security hardening is implemented: SQL injection prevention via prepared statements, password hashing with password_hash() using Bcrypt, XSS prevention via htmlspecialchars() entity encoding on all user-rendered outputs, and secure file upload validation (checking MIME type, size limits under 2MB, and server-side unique filename generation). Accessibility standards (WCAG 2.1 AA) are satisfied with semantic HTML5 landmarks, ARIA attributes, high contrast, and keyboard navigation.

**4. Is project documentation sufficient for installation and testing?**

Yes. The repository includes an exhaustive README.md containing step-by-step local deployment instructions using XAMPP, Apache Virtual Host configuration (<VirtualHost *:80> for studenthub.local with Windows hosts file mapping), complete consolidated MySQL database schema and seed data (database.sql), environment requirements, test credentials for both Admin and Student roles, and verification checklists.

**5. Is the project repository clean and properly committed?**

Yes. The Git repository is cleanly organized with modular directory structures, clear atomic commit messages documenting iterative milestones, sensitive files (.env / temp uploads) excluded via .gitignore, and production-ready source code deployed and tested under Apache/XAMPP.

## B. Implementation / Output

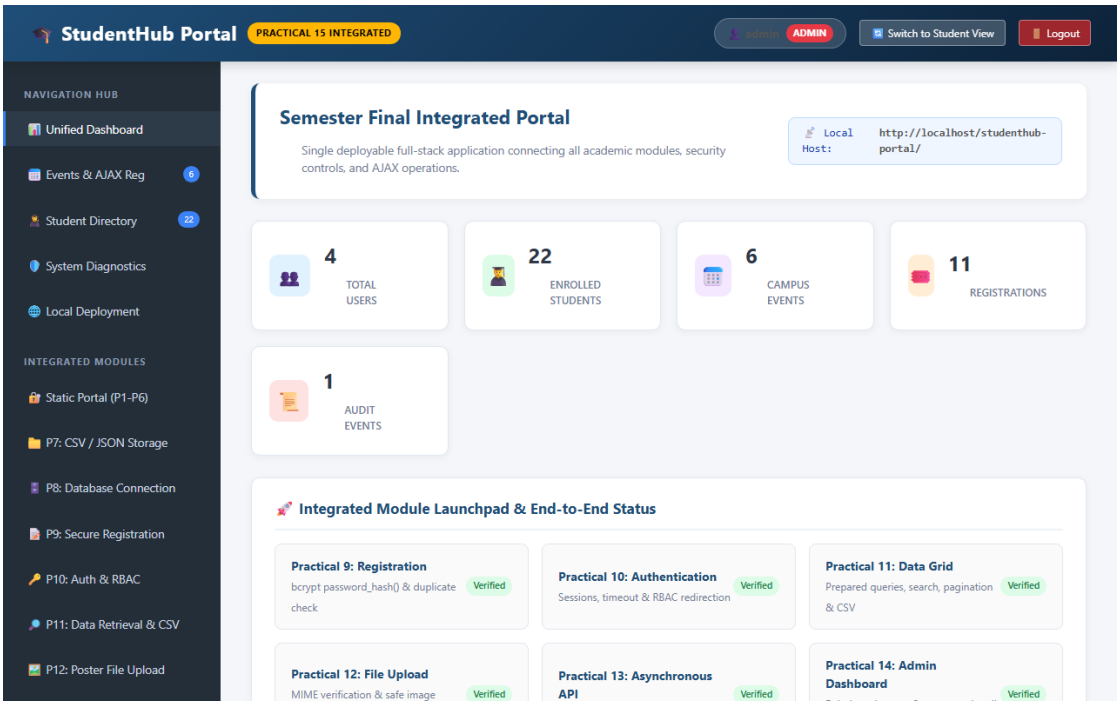



*Figure 15.1: Integrated StudentHub Portal Dashboard (Input Overview, Navigation & Metrics)*

## C. Screenshots

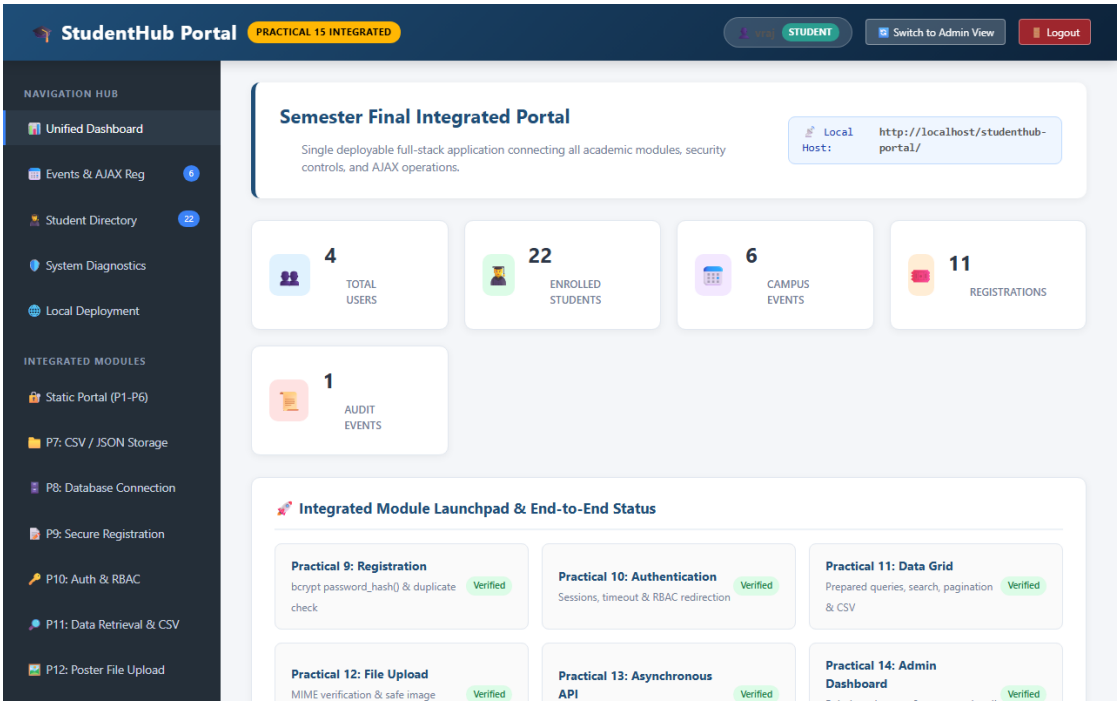



*Figure 15.2: Role-Based Access Control and End-to-End Authentication (Student View & Personalized State)*

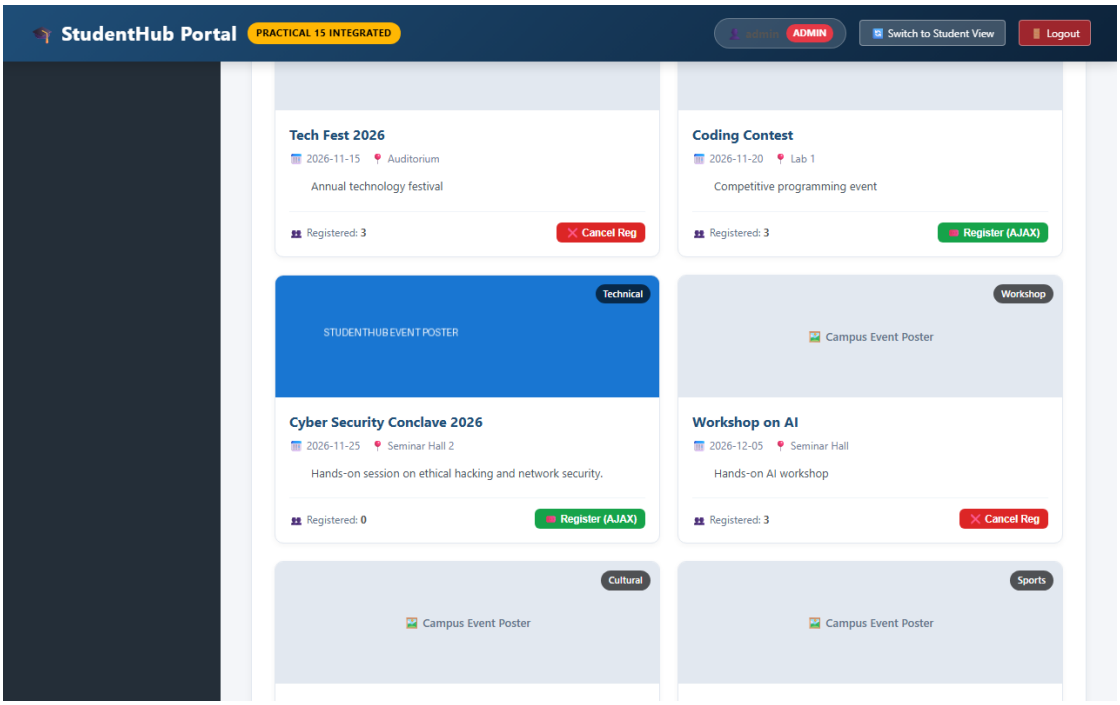



*Figure 15.3: Asynchronous AJAX Operations & Event Registration (Instant Dynamic Registration without Reload)*

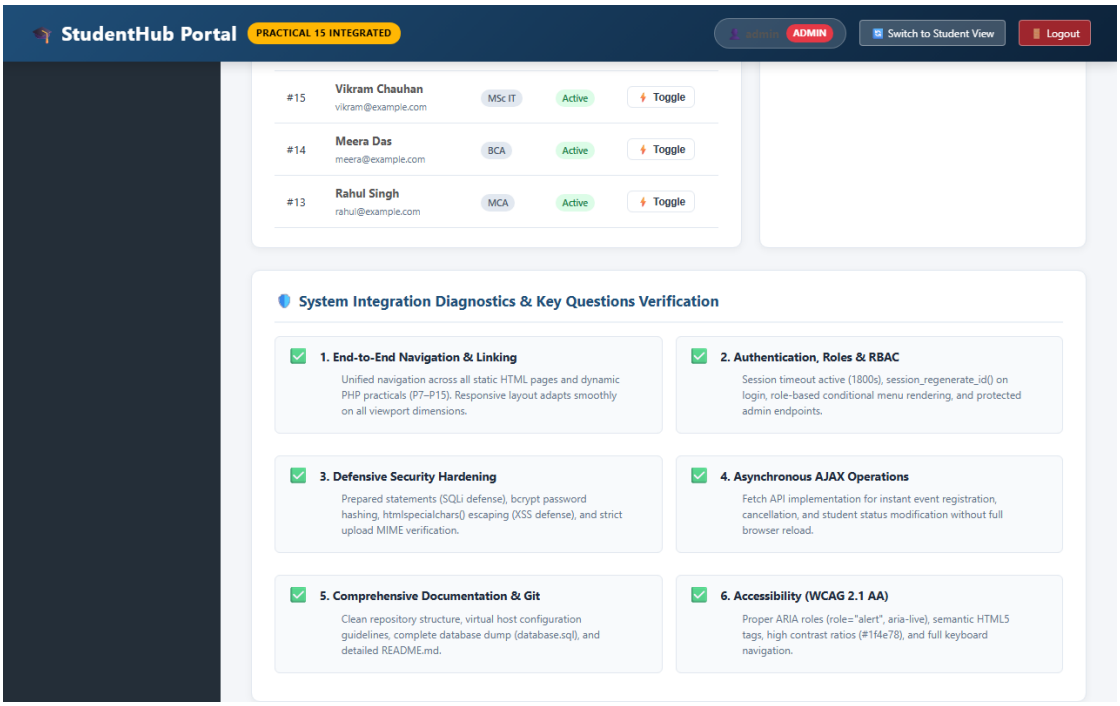



*Figure 15.4: Automated System Diagnostics, Verification Checklist & Apache Virtual Host Deployment (studenthub.local)*

## D. GitHub Link

https://github.com/VRAJ-710/studenthub-portal

## E. Challenges Faced

- Orchestrating seamless integration between heterogeneous components including static HTML pages, PHP session state, MySQL relational tables, and AJAX asynchronous Fetch requests.
- Synchronizing Role-Based Access Control (RBAC) across distinct legacy modules while maintaining unified navigation and preventing privilege escalation.
- Setting up local Apache Virtual Host mapping (studenthub.local) in XAMPP with directory permissions, URL rewrites, and DNS hosts configuration.
- Automating an end-to-end system diagnostic test runner to verify database connectivity, session integrity, security headers, and file upload permissions concurrently.

## F. Learning Outcomes

- Successfully integrated multi-tier web modules into a unified, secure, and production-grade deployable web application.
- Mastered full-stack debugging, automated integration testing, and defensive programming practices (SQLi, XSS, CSRF).
- Gained hands-on expertise in Apache Virtual Host deployment and DNS local resolution in XAMPP.
- Practiced professional Git version control workflows, repository maintenance, and comprehensive technical documentation.

### Faculty Remarks: